

GCP Quota Analysis - Quick Reference Guide

Python Script for Monitoring Quota Usage

The script is located at - https://gitlab.com/gitlab-com/gl-infra/db-migration/-/blob/master/bin/gcs_quota_checker.py

The script pulls current quota usage vs allocation per project and region.

Requirements

```
pip install google-cloud-compute tabulate
```

Basic Usage

```
# Check staging project in us-east1
python gcp_quota_checker.py --project gitlab-staging-1 --region us-east1

# Check all regions
python gcp_quota_checker.py --project gitlab-staging-1 --all-regions

# Filter SSD quotas only (recommended for your case)
python gcp_quota_checker.py --project gitlab-staging-1 --region us-east1 --filter SSD

# Lower threshold to see more quotas
python gcp_quota_checker.py --project gitlab-staging-1 --region us-east1 --threshold 50

# Export to CSV for analysis
python gcp_quota_checker.py --project gitlab-staging-1 --all-regions --output quotas.csv
```

Specific Commands for Your Use Case

1. Check SSD Quotas Before Upgrade

```
# Python version
python gcp_quota_checker.py \
  --project gitlab-staging-1 \
  --region us-east1 \
  --filter "SSD,DISK" \
  --threshold 70 \
  --output ssd_quotas_pre_upgrade.csv
```

2. Check N2D Family Quotas

```
# Python version
python gcp_quota_checker.py \
  --project gitlab-staging-1 \
```

```
--region us-east1 \
--filter "N2D,VM_FAMILY"
```

3. Comprehensive Pre-Upgrade Check

```
# Check all critical quotas across all regions
python gcp_quota_checker.py \
--project gitlab-staging-1 \
--all-regions \
--threshold 60 \
--output pre_upgrade_audit.csv \
--show-all
```

Understanding the Output

Key Metrics to Watch

For PostgreSQL/Patroni Upgrades:

- LOCAL_SSD_TOTAL_GB - Total local SSD across all VMs
- LOCAL_SSD_TOTAL_GB_PER_VM_FAMILY - Per VM family (N2D)
- CPUS - CPU quota for new instances
- PERSISTENT_DISK_SSD_GB - Persistent SSD disks
- IN_USE_ADDRESSES - Internal IP addresses
- INSTANCES - Total VM instances

Example Output

```
High Usage Quotas (>= 70%)
=====
Project          Region    Metric                               Usage    Limit    Usage
gitlab-staging-1 us-east1  LOCAL_SSD_TOTAL_GB                  3600.00  3744.00  96.2%
gitlab-staging-1 us-east1  LOCAL_SSD_TOTAL_GB_N2D              3600.00  3744.00  96.2%
gitlab-staging-1 us-east1  CPUS                                140.00   200.00   70.0%
```

Summary:

```
Total quotas checked: 156
High usage quotas (>= 70%): 3
Critical (>= 90%): 2
Warning (80-90%): 0
```

SSD-related quotas at risk: 2

- gitlab-staging-1/us-east1: LOCAL_SSD_TOTAL_GB = 96.2%
- gitlab-staging-1/us-east1: LOCAL_SSD_TOTAL_GB_N2D = 96.2%

Requesting Quota Increases

If quotas are near limits:

1. Via gcloud CLI

```
# Request quota increase for LOCAL_SSD_TOTAL_GB
gcloud compute project-info describe \
  --project=gitlab-staging-1 \
  --format="value(quotas.filter(metric:LOCAL_SSD_TOTAL_GB))"

# Submit increase request (need to use Console for this)
# Go to: https://console.cloud.google.com/iam-admin/quotas
```

2. Via Console

1. Go to: <https://console.cloud.google.com/iam-admin/quotas>
2. Filter by region: us-east1
3. Filter by metric: LOCAL_SSD_TOTAL_GB
4. Click “EDIT QUOTAS”
5. Request new limit (suggest: +50% above current)

3. Calculate Required Quota

```
# For Patroni v17 upgrade (4 nodes):
# Current per node: 900 GB local SSD (estimated)
# New nodes needed: 4
# Required: 4 × 900 = 3,600 GB
# Plus existing: 3,600 GB
# Total needed: 7,200 GB (current limit: 3,744 GB)

# Request at least: 8,000 GB to have headroom
```

Pre-Upgrade Checklist

Run these commands before upgrade:

```
# 1. Full quota audit
./gcp_quota_check.sh -p gitlab-staging-1 -a -o pre_upgrade_full.csv

# 2. SSD-specific check
./gcp_quota_check.sh -p gitlab-staging-1 -r us-east1 -f SSD -t 60

# 3. CPU check
./gcp_quota_check.sh -p gitlab-staging-1 -r us-east1 -f CPUS -t 60

# 4. Disk check
./gcp_quota_check.sh -p gitlab-staging-1 -r us-east1 -f DISK -t 60

# 5. IP address check
./gcp_quota_check.sh -p gitlab-staging-1 -r us-east1 -f ADDRESS -t 60
```

Automating Quota Monitoring

Cron Job for Daily Checks

```
# Add to crontab
0 8 * * * /path/to/gcp_quota_check.sh -p gitlab-staging-1 -t 80 -o /var/log/quotas/daily_$(date +%Y-%m-%d_%H-%M-%S).log
```

Alert on High Usage

```
#!/bin/bash
# quota_alert.sh

OUTPUT=$(/path/to/gcp_quota_check.sh -p gitlab-staging-1 -r us-east1 -t 80)
EXIT_CODE=$?

if [[ $EXIT_CODE -ne 0 ]]; then
    echo "$OUTPUT" | mail -s "GCP Quota Alert: High Usage Detected" ops@gitlab.com
fi
```

Troubleshooting

“Permission Denied” Error

```
# Ensure you have compute.regions.get permission
gcloud projects get-iam-policy gitlab-staging-1 \
  --flatten="bindings[].members" \
  --filter="bindings.members:user:$(gcloud config get-value account)"
```

“Project Not Found” Error

```
# List accessible projects
gcloud projects list

# Set active project
gcloud config set project gitlab-staging-1
```

Quotas Not Updating

```
# Clear gcloud cache
gcloud compute regions describe us-east1 \
  --project=gitlab-staging-1 \
  --format=json > /dev/null

# Wait 1-2 minutes and retry
```

Additional Resources

- [GCP Quotas Documentation](#)

- Request Quota Increase
- Understanding VM Families
- Local SSD Best Practices

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